# Sub-module 5: Marketing and Supply Chain for Managers Inventory Management

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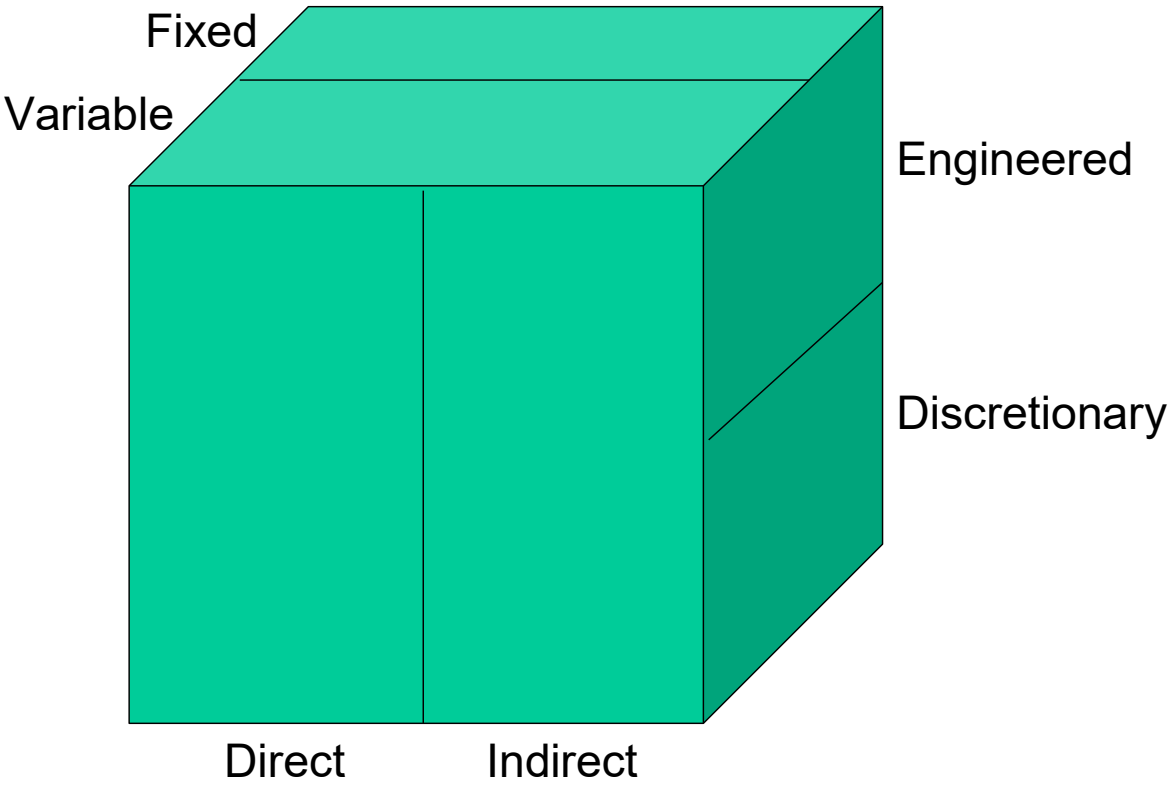
# Readings

Lecture slides

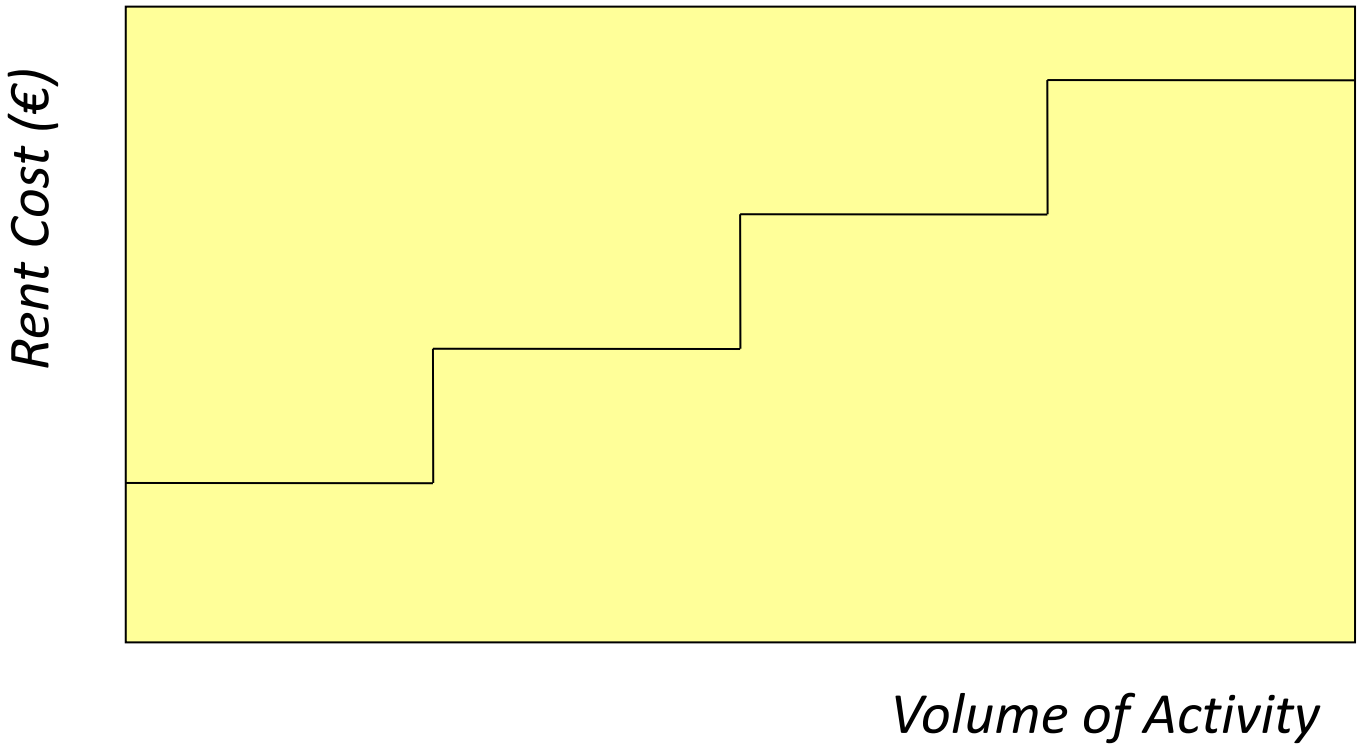
Glossary

Panigrahi, R. R., Shrivastava, A. K., & Nudurupati, S. S. (2024). Impact of inventory management on SME performance: a systematic review. *International Journal of Productivity and Performance Management*, *73*(9), 2901–2925. https://doi.org/10.1108/IJPPM-08-2023-0428

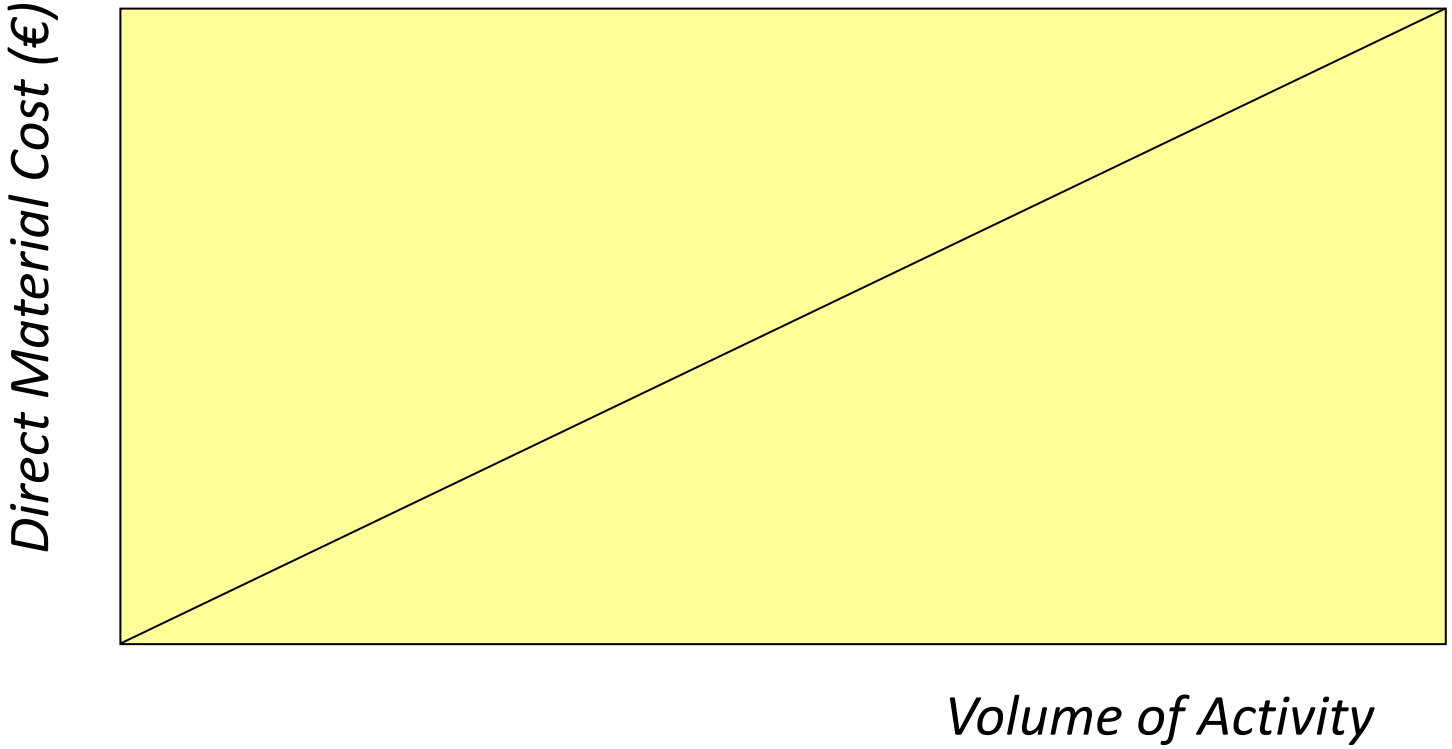
# Total Cost


Fixed
Variable
Engineered
Discretionary
Direct
Indirect

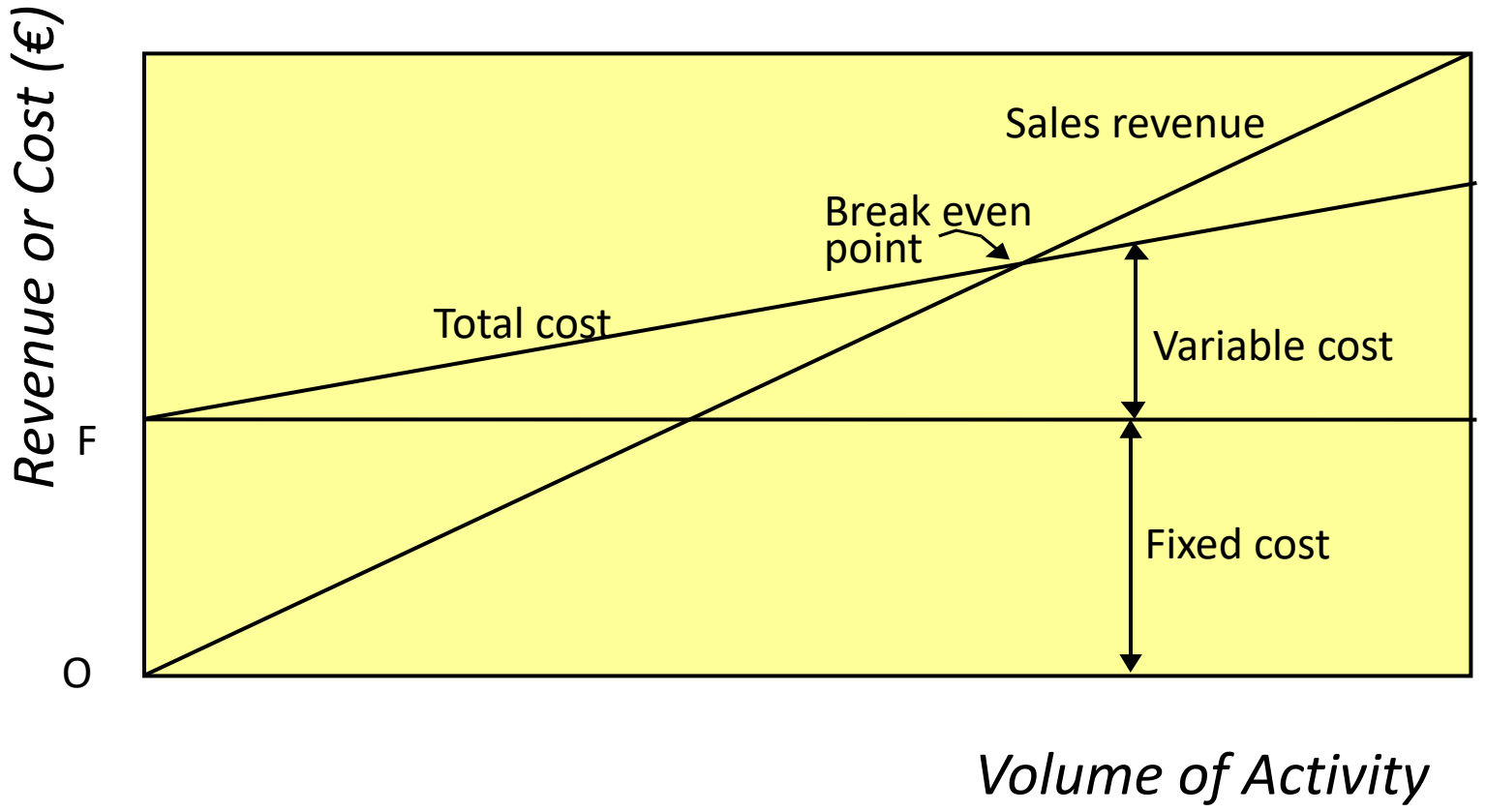
# Rent Cost vs Volumee of Activity



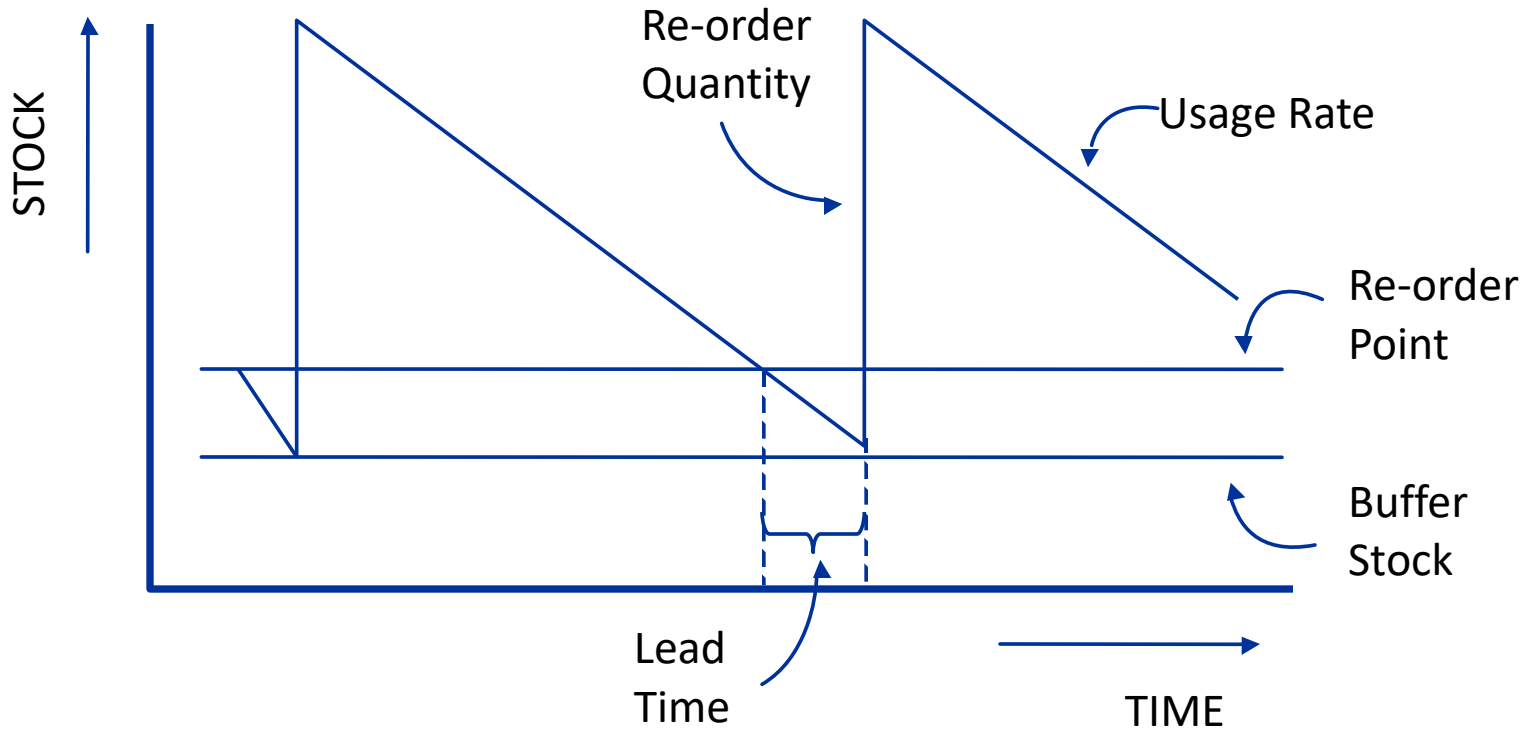
# Direct Cost vx Volume of Activity



# Break-Even Chart



# When: the re-order point




Notes:

1. Re-order Point = demand during lead time + safety stock
2. Re-order quantity = Economic Order Quantity
3. Buffer Stock = *f*(service level. lead time variability.demand variability)

# JIT Key areas

| **In-Company tools** | **Inter-company tools** | **Supporting mechanisms** |
|---|---|---|
| Modular designs | Certified quality | Quality management |
| Focus | Supplier reduction | Value engineering |
| Total productive maintenance | Joint development programmes | Time based process mapping |
| Setup reduction | JIT delivery | Six sigma |
| Layout and flow | EDI + Internet | Methods analysis |
| Pull scheduling | Integrated transport | Error proofing |
| Small machines | Standard containers | Management by sight |
| Lot size reduction | Bar coding/RFID's | Enforced problem solving |
| JIT/MRP | Long term contracts | Under-capacity scheduling |

# Some trends in inventory management

- Fewer but larger inventories
- High degree of automatisation
- Datadriven warehouses
- Real-time data analytics

- Please remember, the only way to replace an inventory is with information.



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